WEEK:1

Aim: To implement data preprocessing by identifying and handling missing values in the Iris dataset using the Hugging Face Dataset library and Mean Imputation technique.

Introduction:
In machine learning and deep learning, the quality of data plays an important role in model performance.In Real-world datasets often contain incomplete data in the form of missing values.Deep learning models cannot directly process missing values because mathematical operations require complete numerical inputs.So, data preprocessing is performed before training a model.In this experiment,we have taken the Iris dataset and Some missing values are introduced into the dataset, detected, and replaced using the Mean Imputation technique with the help of Hugging Face Dataset processing.

The Iris dataset is a commonly used machine learning dataset that contains information about iris flowers.

The dataset contains:

- 150 records (samples)
- 4 numerical features
- 1 target class

CODE:

```
from datasets import Dataset
from sklearn.datasets import load_iris
import pandas as pd
import numpy as np

iris = load_iris()
df = pd.DataFrame(
   iris.data,
   columns=[
       "sepal_length",
       "sepal_width",
       "petal_length",
       "petal_width"
   ]
)
df["species"] = iris.target
```

```
print("Original Iris Dataset:")
print(df.head())

df.loc[2, "sepal_length"] = np.nan
df.loc[5, "petal_width"] = np.nan
df.loc[10, "sepal_width"] = np.nan
df.loc[20, "petal_length"] = np.nan

print("\nDataset with Missing Values:")
print(df.head(25))

print("\nMissing Values Count:")
print(df.isnull().sum())

dataset = Dataset.from_pandas(df)

print("\nHugging Face Dataset:")
print(dataset)
def fill_missing_values(example):
    example["sepal_length"] = (
        df["sepal_length"].mean()
        if example["sepal_length"] is None
        else example["sepal_length"]
    )

    example["sepal_width"] = (
        df["sepal_width"].mean()
        if example["sepal_width"] is None
        else example["sepal_width"]
    )

    example["petal_length"] = (
        df["petal_length"].mean()
        if example["petal_length"] is None
        else example["petal_length"]
    )

    example["petal_width"] = (
        df["petal_width"].mean()
        if example["petal_width"] is None
        else example["petal_width"]
    )
```

```
    return example

clean_dataset = dataset.map(fill_missing_values)
print("\nDataset After Handling Missing Values:")
print(clean_dataset)
```

OUTPUT:

Original Iris Dataset:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

Dataset with Missing Values:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	NaN	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0
5	5.4	3.9	1.7	NaN	0
6	4.6	3.4	1.4	0.3	0
7	5.0	3.4	1.5	0.2	0
8	4.4	2.9	1.4	0.2	0
9	4.9	3.1	1.5	0.1	0
10	5.4	NaN	1.5	0.2	0
11	4.8	3.4	1.6	0.2	0
12	4.8	3.0	1.4	0.1	0
13	4.3	3.0	1.1	0.1	0
14	5.8	4.0	1.2	0.2	0
15	5.7	4.4	1.5	0.4	0
16	5.4	3.9	1.3	0.4	0
17	5.1	3.5	1.4	0.3	0
18	5.7	3.8	1.7	0.3	0
19	5.1	3.8	1.5	0.3	0
20	5.4	3.4	NaN	0.2	0
21	5.1	3.7	1.5	0.4	0
22	4.6	3.6	1.0	0.2	0

```
23          5.1         3.3          1.7         0.5        0
24          4.8         3.4          1.9         0.2        0

Missing Values Count:
sepal_length    1
sepal_width     1
petal_length    1
petal_width     1
species         0
dtype: int64

Hugging Face Dataset:
Dataset({
    features: ['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
'species'],
    num_rows: 150
})

Map:  100%
  150/150  [00:00<00:00,  3214.03  examples/s]
Dataset After Handling Missing Values:
Dataset({
    features: ['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
'species'],
    num_rows: 150
})
```